

True Stage One Technical Report

Environment validation, baseline discrimination, instrumentation, and frozen evidence

Core Research Document - Implemented evidence report

Decentralized Swarm Emergence Research Program

Stage One revision - September 2026

Evidence status: Stage One validates the experimental environment, instrumentation, reward accounting, heuristic baselines, and reproducibility pipeline. It does not yet establish learned emergence, a critical alpha threshold, learned communication, or a phase transition.

Executive summary

True Stage One converts the original swarm concept into a reproducible experimental laboratory. The completed stage validates environment semantics, anonymous partial observations, reward accounting, baseline policies, deterministic collection, telemetry, visualization, and artifact generation. The frozen dataset contains 200 policy/reward conditions. Structured heuristics strongly separate from the random control, while alpha leaves heuristic trajectories invariant as expected because those policies do not learn from reward. These results validate the measurement substrate; they do not establish learned emergence.

1. Implemented system

Component	Implemented role
SwarmWorld	deterministic grid physics and event attribution
PettingZoo ParallelEnv	multi-agent lifecycle, observations, rewards
Baseline policies	random, greedy, flocking, distributed-greedy oracle
Experiment collector	episodes, metrics, sweeps, provenance
Visualization	behavioral charts, trajectory overlays, visitation heatmaps
Live laboratory	interactive policy/reward/seed/swarm controls and export

2. Observation and decentralization contract

Each agent receives a fixed tensor of shape $(3, 2r+1, 2r+1)$ centered on itself. Channels encode ego position, nearby-agent occupancy, and nearby-target occupancy. Absolute position and agent identity are omitted. This creates a stable shared-policy interface while preserving the capstone's partial-observability requirement.

3. Reward accounting correction

The implemented hybrid reward assigns literal local collection credit at $\alpha = 0$ and fully shared collection credit at $\alpha = 1$. This corrects a legacy failure mode in which every local agent could receive a team-level step collection count, which would have contaminated the intended local/global comparison.

4. Frozen evidence

Policy	Success rate	Completion rate	Steps	Distance	Collision rate
Random	0.10	0.70	174.8	787.7	0.0086
Greedy	1.00	1.00	13.5	81.0	0.0659

Flocking	1.00	1.00	16.5	90.9	0.1109
Distributed-greedy oracle	1.00	1.00	34.9	87.4	0.0637

At $\alpha = 0.5$, ten independent seeds were used for the repository's greedy-versus-random Welch comparison. Greedy completion exceeded random completion ($t = -4.609$, $p = 0.00127$). The statistical result demonstrates that the task/measurement stack distinguishes structured heuristic behavior from the stochastic control. It is not evidence that a learned swarm has emerged.

5. Alpha invariance as a validation check

The frozen matrix is four policies x five alpha settings x ten seeds. Heuristic actions are not optimized against reward, so alpha should alter calculated return while leaving same-policy, same-seed trajectories unchanged. Observed behavioral invariance across alpha is therefore a validation property, not a null result about reinforcement learning.

6. What Stage One establishes

- Repeatable seeded environment execution.
- Measurably different behavior across baseline policies.
- Complete performance and behavioral telemetry.
- Reward-topology accounting that does not silently alter fixed-policy behavior.
- An experimental interface ready for matched Shared PPO and Independent PPO training.

7. What Stage One does not establish

- No trained PPO policy has been evaluated.
- No alpha-dependent learned behavioral transition has been measured.
- No critical α^* has been estimated.
- No learned communication or communication-budget threshold has been tested.
- No large-swarm scalability or failure-resilience claim is supported.

8. Threats to validity

- The reference task is a discrete foraging environment; conclusions may be task-specific.
- Several strong heuristics solve the current reference layouts, potentially creating ceiling effects for raw success rate.
- The oracle baseline uses global target assignment and must never be represented as a decentralized policy.
- Behavioral metrics such as role differentiation require careful interpretation because apparent specialization can arise from geometry or chance.
- Ten heuristic seeds validate baseline separation but do not determine the eventual number of independent RL training runs required.

9. Next experiment

The next stage trains Shared-policy PPO and Independent PPO under identical budgets across the pre-registered alpha sweep, evaluates them on held-out episode seeds, and compares them with the frozen baselines. Only after those data exist should the project estimate α^* , discuss an emergence curve, or test communication.

10. Reproducibility checklist

- Record code revision and configuration in each evidence manifest.
- Separate training seeds from evaluation seeds.
- Preserve raw episode-level telemetry and seed-level summaries.
- Retain failed/poor-performing seeds unless a documented software defect invalidates the run.
- Generate figures from the same frozen data used for statistical analysis.
- Version protocol amendments before interpreting affected results.

Figures

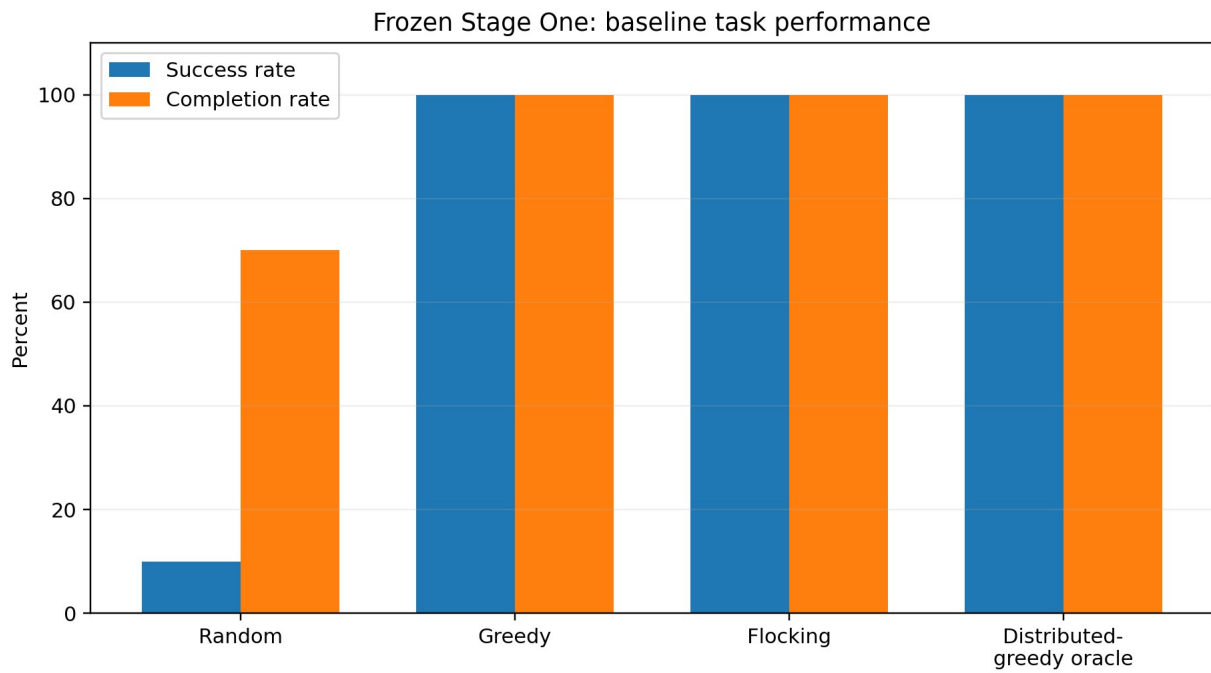


Figure 1. Frozen Stage One task performance by baseline policy.

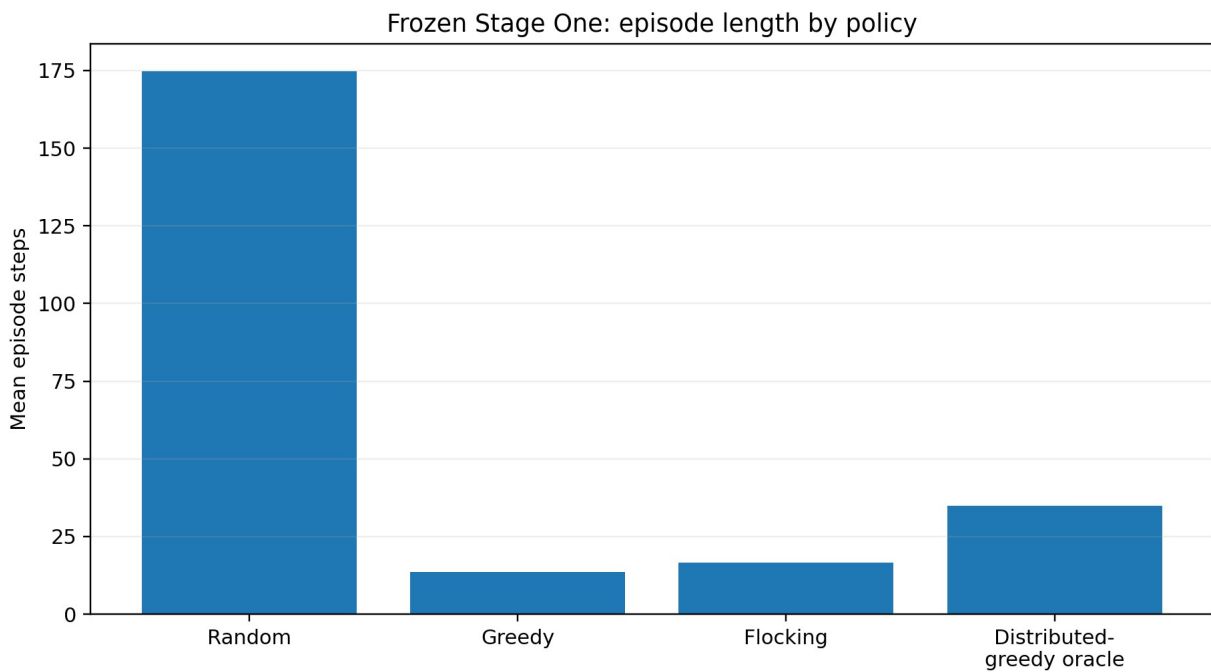


Figure 2. Frozen Stage One mean episode length by baseline policy.

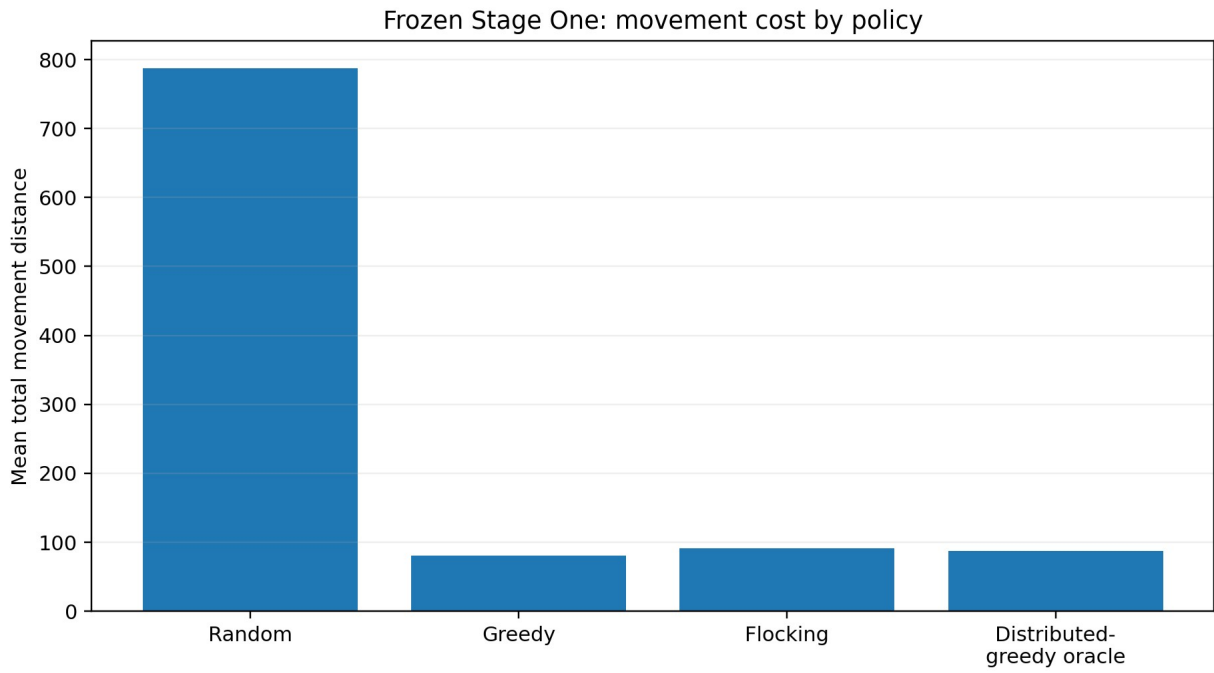


Figure 3. Frozen Stage One mean movement cost by baseline policy.

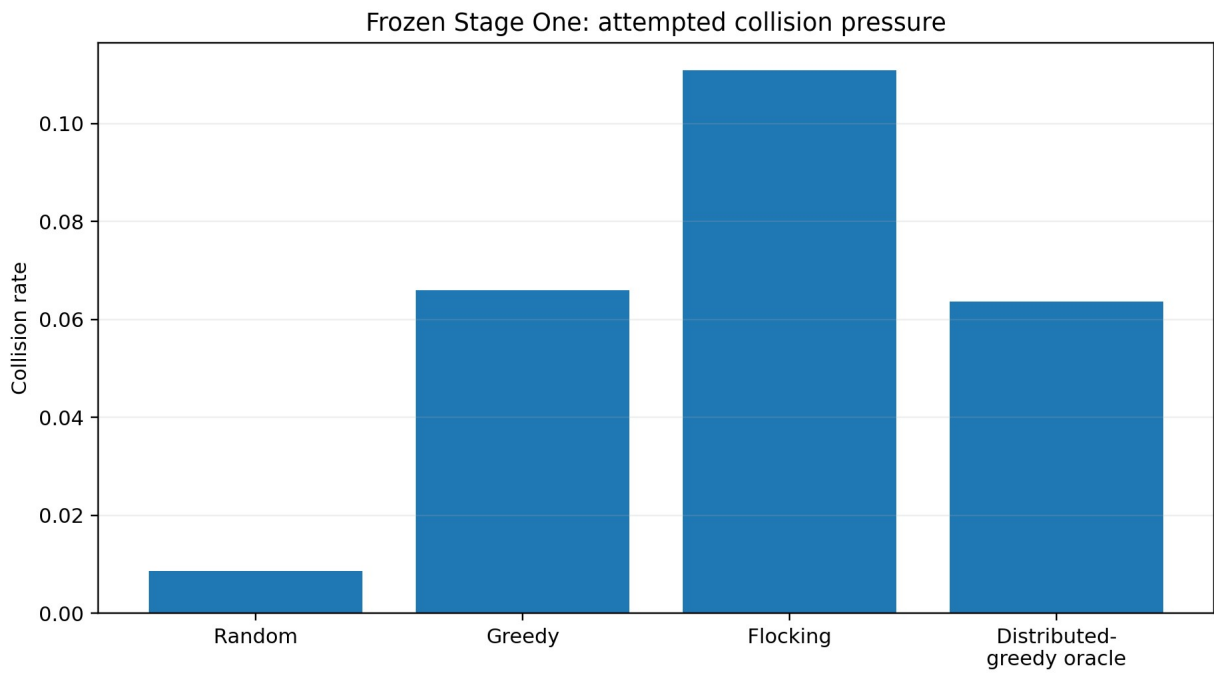


Figure 4. Frozen Stage One attempted collision rate by baseline policy.